

UNIVERSITY FACULTY OF ENGINEERING

MODULE: MATH201 - ENGINEERING CALCULUS & ODEs

TIME: 2 HOURS | TOTAL MARKS: 50

QUESTION 1: Damped Mechanical & Circuit Oscillations [25 Marks]

A mechanical spring-mass-damper system is modeled by the second-order differential equation:

$$y''(t) + 4y'(t) + 13y(t) = 0$$

Subject to initial conditions: $y(0) = 2$ and $y'(0) = -1$.

- (a) Formulate the characteristic auxiliary equation and solve for the roots. [5 Marks]
- (b) Write down the general solution for the displacement $y(t)$. [8 Marks]
- (c) Determine the exact particular solution satisfying initial conditions. [8 Marks]
- (d) Classify whether the physical system is overdamped, critically damped, or underdamped. [4 Marks]

QUESTION 2: Multivariable Integration & Flux [25 Marks]

Evaluate the line integral along the parabolic curve C:

$$\text{Integral over } C \text{ of } (2xy \, dx + (x^2 + y^2) \, dy)$$

from the point $(0, 0)$ to $(1, 1)$. Verify if the vector field is conservative.